



RETAINIQ Customer Intelligence Engine

End-to-End Enterprise MLOps Pipeline & Real-Time Churn Prevention

“Transitioning from Reactive Churn Discovery to Autonomous Retention Intervention”



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COHORT PROGRAM
G1 Data Analytics

INSTITUTION
SURE ProEd

INTEGRATED TECHNOLOGY STACK



Python



SQL



Scikit-Learn



Apache Airflow



Streamlit



Microsoft Power BI

The Retention Challenge: Moving from Reactive to Proactive

5x–7x Ratio

Acquiring a new customer costs 5x–7x more than retaining an existing one.

⚠️ Traditional Legacy Approach

- ✗ Monthly CSV exports with delayed reporting cycles.
- ✗ Churn discovered after customers and revenue loss.
- ✗ Static spreadsheets with zero behavioral scoring.
- ✗ Manual, sporadic email follow-ups by sales reps.

Result: Unpredictable revenue loss and reactive firefighting.

✅ RetainIQ Autonomous Engine

- ✓ Autonomous daily batch ingestion into SQL database.
- ✓ Proactive ML prediction before customers leave.
- ✓ 11 Behavioral RFM loyalty customer personas.
- ✓ Automated daily Gmail alerts dispatched to team inboxes.

Result: Continuous retention and automated customer outreach.

Six Integrated Engineering Components

1. Relational Database

SQLAlchemy + SQL

Structured database maintaining integrity for 5,630 customer accounts.

2. RFM Segmentation

Pandas Analytics Engine

Quantifies Recency, Frequency, and Monetary loyalty dimensions.

3. Machine Learning Core

Tuned Random Forest

Trained to detect early churn signals with 81.8% accuracy.

4. Workflow Orchestration

Apache Airflow 2.x

Autonomous daily pipeline scheduling with automatic error recovery.

5. Operations Hub

Streamlit Web Application

Visual interface for sales outreach, batch simulation, and manual review.

6. Executive BI

Microsoft Power BI

Interactive 1-page Control Tower providing instant visual analytics.

Operational Pipeline Flow & Live Operations Console

1 Daily SQL Ingest



2 RFM Scoring



3 ML Prediction



4 Gmail Alerting



5 Power BI Refresh



The Streamlit Operations Console

Provides non-technical managers with an interactive control hub to monitor portfolio KPIs, filter the high-risk customer queue, dispatch retention emails, and simulate incoming daily batches on demand.



How Power BI Refreshes Seamlessly

The pipeline pre-calculates and exports clean analytical summary CSVs. When clicking Refresh, Power BI re-reads these flat files directly into memory without database locks or query lag.



Live Demonstration: Switching to Streamlit Operations Console & Power BI Dashboard.

Proven Dataset Metrics & Root-Cause Insights

5,630

Total Customer Accounts
Monitored in SQL

86.31%

Baseline Retention Rate (13.69%
Churn Rate)

339

High-Risk Accounts (Exceeding ≥
80% Risk)

81.8%

Machine Learning Classification
Accuracy

Product Category Churn Rates

Grocery (Leading Attrition)	16.0%
Laptop & Accessory	13.9%
Others & Household	13.8%
Fashion & Apparel	13.5%
Mobile Phone (Most Loyal Category)	11.4%

Customer Tenure Danger Zone

Months 0 to 6 Onboarding Cliff

Churn risk peaks above 40% for new accounts during early onboarding.

Steep Post-10 Month Risk Drop

Risk stabilizes around 10%–15% once accounts pass month 10.